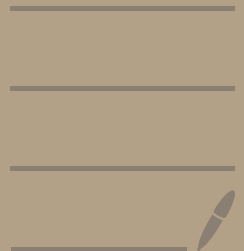


• Lec 10. • Extending solution  
• Blow up



- (IV) Extensibility of solution + Blow-up.  
Does a solution exist for all time?

Ex 
$$\begin{cases} \frac{d}{dt} y = y^2 \\ y(0) = 1 \end{cases} \Rightarrow y(t) = \frac{1}{1-t}$$

$y(t) \rightarrow \infty$  as  $t \rightarrow 1^-$ . Blow-up!

• Consider

(i) 
$$\dot{x} = f(t, x).$$

Ass 1: we assume  $f \in C(\mathbb{R}, \mathbb{R})$  continuous every

Ass 2:  $f$  is Lipschitz in any compact region. e.g.

$$|f(t, x) - f(t, y)| \leq L_0 |x - y|, \quad \forall x, y \in B$$

Lemma (Gluing). Suppose that  $\phi_i(t)$  is a  $C^1$  solution to ① defined on  $t \in I_i = (a_i, b_i)$  and  $I_1 \cap I_2 \neq \emptyset$ .

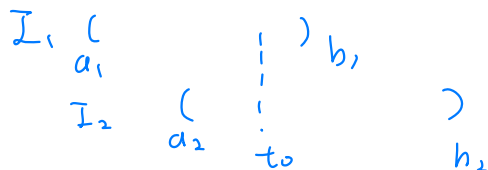
• suppose that  $t_0 \in I_1 \cap I_2$

with  $\phi_1(t_0) = \phi_2(t_0) = x_0$

Then. 1)  $\phi_1(t) = \phi_2(t)$ ,  $\forall t \in I_1 \cap I_2$

$$2) \phi(t) = \begin{cases} \phi_1(t) & t \in I_1 \\ \phi_2(t) & t \in I_2 \end{cases}$$

is a  $C^1$  solution to ① for  $t \in I_1 \cup I_2$ ,  
and  $\phi(t_0) = x_0$ .



Pf: 1). Since  $\phi_1(t), \phi_2(t)$  are 2 solutions to ① on  $t \in I_1 \cap I_2$  with  $\phi_1(t_0) = \phi_2(t_0)$ .

By existence and uniqueness theorem, we get.

$$\phi_1(t) = \phi_2(t), \quad t \in I_1 \cap I_2.$$

Rigorous version. Need to define

$$J = \{t \in I_1 \cap I_2 : \phi_1(t) = \phi_2(t)\}$$

$t_0 \in J$ .  $J$  is relative open (By existence + Uniqueness)  
is relative closed (By continuity).

$$\Rightarrow J = I_1 \cap I_2$$



(2) Now, from 1),  $\phi(t)$  is well defined,  $C^1$  solution to ODE ①.  $\phi(t_0) = \phi_1(t_0)$  □

Thm [Extension].  
: (Similar to Lem 2.14. But easier).

Let  $\phi(t)$  be a solution to ① defined on interval  $(t_-, t_+)$ . Then there exists an extension to interval  $(t_-, t_+ + \varepsilon)$  if and only if

(A)  $\lim_{t \rightarrow t_+^-} |\phi(t)| < +\infty$ .  
*(left limit)*

(B) There exists a sequence of  $t_n \rightarrow t_+^-$  s.t.  
 $\phi(t_n) \rightarrow \phi_0$  with  $|\phi_0| < +\infty$

Fact: Conds (A)  $\Leftrightarrow$  (B) !

Rmk: The Thm: cannot extend beyond  $t_+$   
 $\Leftrightarrow \lim_{t \rightarrow t_+^-} |\phi(t)| = +\infty$

Pf: " $\Rightarrow$ " extend  $\Rightarrow$  limit.

If  $\phi(t)$  extends to  $(t_-, t_+ + \varepsilon)$ . Since  $\phi(t)$  is  $C^1$   $(t_-, t_+ + \varepsilon)$ , then

$$\lim_{t \rightarrow t_+^-} \phi(t) = \phi(t_+), \text{ bounded.}$$

" $\Leftarrow$ " limit  $\Rightarrow$  extend.

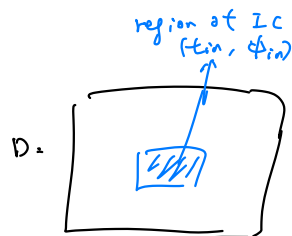
Suppose  $\phi(t_n) \rightarrow \phi_0$ ,  $|\phi_0| < \infty$ .  
 $n \rightarrow \infty$ .

Then  $\exists N$  large s.t.  $n > N$ .

$$\phi(t_n) \in [\phi_0 - 1, \phi_0 + 1]$$

Step 1: consider region

$$D = [t_+ - 2, t_+ + 2] \times [\phi_0 - 2, \phi_0 + 2]$$



Since  $f \in C(D)$   $|f| \leq M$   $(t, y) \in D$

$$|f(t, y_1) - f(t, y_2)| \leq L, \quad (t, y_i) \in D$$

Step 2:  $\forall t_n \in [t_+ - 1, t_+ + 1], \phi_{in} \in [\phi_0 - 1, \phi_0 + 1]$

By existence & uniqueness, we can solve (1)

$$\text{for } h \geq \min\left(\frac{1}{2L}, \frac{2}{M}, 1\right) > 0.$$

Step 3: Take  $t_m$ , s.t.  $t_m + h > t_+$

solve ODE with  $\begin{cases} \phi_{in} = \phi(t_m) \\ t_{in} = t_m \end{cases}$

$\Rightarrow$  get new solution  $\phi_{new}(t)$  defined on  $[t_m - h, t_m + h]$

$$\phi_{new}(t_m) = \phi(t_m).$$

By Gluing Lemma  $\Rightarrow$

Extend  $\phi(t)$  to  $[t_m - h, t_m + h] \cup (t_-, t_+)$   
 $\supseteq [t_-, t_+ + \varepsilon)$

• App:

Let  $w$ ).

①  $\frac{d}{dt} y = f(t, y)$ ,  $0 \leq |f| \leq M$ ,

②  $\partial_y f \in C(\mathbb{R} \times \mathbb{R})$

$\Rightarrow$  solutions exist everywhere.

②  $\frac{d}{dt} y = y f(t, y)$

$|f| \leq M$

$\Rightarrow$  Global soln.

$\partial_y f \in C(\mathbb{R} \times \mathbb{R})$

③  $\frac{d}{dt} y = \frac{e^{-y^2} \cdot \sin t}{1+t^2+y}$

improvement

but

$\left| \frac{d}{dt} y \right| \leq 1 \Rightarrow$  Global solution.

- Finite time blowup, [Use comparison].  
How can we quickly determine a solution does not exist for all time?

- Canonical example. Riccati eqn

$$\begin{cases} \frac{d}{dt} y = y^p, & p > 1. \\ y|_{t=0} > 0. \end{cases} \quad \begin{array}{l} \text{i.e. result ③} \\ \text{is sharp.} \end{array}$$

- Apply comparison principle to analyze.

- $\frac{d}{dt} y = e^y$

- $\frac{d}{dt} y = y^2 - cy$

- $\frac{d}{dt} y = e^{-y}$   
 $\begin{cases} y(0) > 0 \end{cases}$

• Time allow. consider behavior of [optional  
this time]

$$\dot{y} = y^2 + b y + c$$

discuss behavior of h.c. with different I.C.